Theoretical Tasks

Task 2.1 Ethical Assessment

Theoretical Background

Models such as GPT-4, Stable Diffusion, and Copilot are impressive in their ability to generate realistic images but have limitations. These models require an extensive amount of training data that can be difficult to obtain and may not fully represent the complexity of the world we live in. Therefore, they may generate images that are visually convincing but contain inaccuracies or inconsistencies with reality.

While these models are undeniably useful for certain purposes, it is essential to remember that they have limitations. The models should not be solely relied upon as a complete representation of reality. It is necessary to consider their constraints and limitations while using them for practical applications. Therefore, performing a thorough analysis and validation before implementing these models in real-world scenarios is crucial. By doing so, we can ensure that these models are helping us to achieve our goals without compromising their reliability and accuracy.

Here, you will use any of the generative image LLMs (GPT-4, Copilot, Gemini, etc.) to generate pictures of humans in real-life scenarios. The main goal is that you asses the images generated (at least 5) and carefully check for inconsistencies.

Present your findings (writing) with examples (optional). You should be able to find one inconsistency with reality and present that, explaining why you think it happened.

Hint: Check their body composition!


SUMMERFEST '24
LIVE MUSIC


SÖRHÁZ A BÚSULÓ JUHHOZ
CSAPOLT SÖRÖK
NYITVA: 10:00-02:00
BÚSULÓ JUH KOCSMA


KECSKEMÉT, DOBÓ KRT. 12.
DOBÓ ISTVÁN KÖRÚT
12
DÉLI PIZZA
100% FRISS & FINOM
KEC-1234


HAPPY NEW YEAR 2024
TIMES SQUARE
HAPPY NEW YEAR 2024
CELEBRATE
HAPPY NEW YEAR 2024


CSONAKKSEU-PARK,
RÉCSNEMÉT

The first thing I noticed that if I do not give details to the AI and just ask to generate an image with people on it, it will generate an image where all of the people are happy and they are in a good situation.

Size issue:

There are also some issues with the size for example on the third image with pizza delivery man, he has a small carrier but the pizzas would not fit into it and also he would not even need that carrier because he already has a carrier on the scooter. On the 4th image also has some issues with the sizes the worm is too big this is not so realistic.

Color and quality issue:

The colors are always really good and the images are too colorful. Quality is really good on every image it might better than the pictures that a professional photographer could take.

Crowd issue:

On the first picture in the crowd there is a repeating pattern almost all of the people have one hand up and the other is down no one put both of their hands up and also the facial expressions are also the same all of the people are happy and have their mouth open.

Little issues:

On the 7th image the photo in the lady's hand is transparent but we should just see the back of the photo. On the 6th image the „Happy new year" titles are not correct and blurry in the background.

Number of fingers

On the first and second images there are some issues woth the fingers.

Reason:

Ai does not have anatomical knowledge. The model rest on statistics and probabilities of pixels. It knows that there should be skin colored pixels at the place of our hands but there are a lot of hand gestures where not all of the fingers are visible so Ai just get confuse and simply place finger-like shapes. As we can see on the second image that I have just paste here it almost look like a „like" hand signal, it just did not generate a finger.